

PEDRO PABLO ZAVALA TEJOS

Computer Science Engineer – Pontificia Universidad Católica de Chile

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EDUCATION

Pontificia Universidad Católica de Chile

Computer Science Engineering

2021 - Jul. 2026
Santiago, Chile

- Major in Computing and Information Systems.
- Minor in Data Science and Analytics.
- Academic average: 6.22/7.00.

PROFESSIONAL EXPERIENCE

Research Assistant

School of Engineering UC

Mar. 2025 - Present
Santiago / Estación Patagonia UC

- Contribute to SAVIIA and ECHO, a hybrid edge-to-cloud data infrastructure for UC's Regional Field Stations Network, supporting reliable acquisition, validation, synchronization, monitoring, and governance of scientific and operational data.
- Developed ECHO, a Home Assistant integration for orchestrating data-acquisition workflows, sensor monitoring, automations, and local IoT devices, and EchoLib, a Python library for heterogeneous sources including FTP/SFTP, REST APIs, sensors, and cloud storage.
- Design reproducible ETL pipelines, backend services, and APIs for PISMA-WP, integrating satellite, geospatial, environmental, and field data with metadata, traceability, versioning, and quality-control mechanisms.
- Develop and evaluate computer-vision workflows for camera-trap imagery, including dataset preparation, animal detection, species classification, transfer learning, model evaluation, and edge-inference experimentation.

Undergraduate Researcher - SCORPIO

IoT-UC Laboratory, School of Engineering UC

Mar. 2026 - Jun. 2026
Santiago, Chile

- Designed and implemented an edge-to-cloud architecture to acquire, process, store, transmit, and visualize LoRa satellite data using containerized edge services, MQTT, SQLite, a REST API, and a geospatial web interface.
- Designed the system for experimental traceability and continued operation under intermittent connectivity, providing a foundation for distributed satellite signal-measurement stations.

Capstone Project - Intelligent Data Segmentation Assistant using Natural Language

Mar. 2026 - Jun. 2026

National Center for Artificial Intelligence (CENIA), School of Engineering UC Santiago, Chile

- Developed, as part of a six-member team, a conversational assistant that translates natural-language requests into validated and executable SQL queries.
- Contributed to a cloud-based architecture integrating Slack, RAG, PostgreSQL with pgvector, LLM-based SQL generation, validation, user feedback, and automated deployment using AWS and Terraform.

Big Data Engineer Intern

SimplePark

Jan. 2025 - Apr. 2025
Remote

- Developed and automated data extraction, transformation, and analytics workflows using Google Cloud services and BigQuery.
- Optimized NoSQL data models and processing workflows for operational datasets used in analysis and decision-making.

Capstone Project - AI-powered web platform for academic management at the UC School of Health Sciences.

School of Health Sciences, Pontificia Universidad Católica de Chile

Aug. 2024 - Dec. 2024
Santiago, Chile

- Implemented backend services for an AI-assisted web platform using NestJS, TypeScript, Prisma, and PostgreSQL.
- Integrated external data sources and redesigned relational schemas to improve maintainability and scalability.

Teaching Assistant - Advanced Computer Programming (IIC2233)

Pontificia Universidad Católica de Chile

Mar. 2023 - Jul. 2024
Santiago, Chile

- Guided students during programming sessions and evaluated Python assignments and assessed activities.

PROFESSIONAL SKILLS

Programming: Python, C/C+, TypeScript/JavaScript, SQL

Backend and Data: FastAPI, NestJS, REST APIs, PostgreSQL, BigQuery, MongoDB, Apache Spark, ETL pipelines

Edge and IoT: MQTT, Home Assistant, FTP/SFTP, sensors, embedded devices

Cloud and Tools: Google Cloud Platform, AWS, Docker, Git/GitHub

EXTRACURRICULAR ACTIVITIES

Presenter

ECHO: Monitoring and Automation Platform for Remote Stations

Congreso Ingeniería para Chile PUC
May 2026

Private Mathematics Tutor

Jan. 2024 - Jan. 2025

LANGUAGES

Spanish: Native

English: B2 / Professional working proficiency

ADDITIONAL INFORMATION

Portfolio

pedrozavalat.github.io

LinkedIn

linkedin.com/in/pedro-pablo-zavala

GitHub

github.com/pedrozavalat

REFERENCES

Rodrigo Arnaldo Carrasco Schmidt

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